Title : Hospital Management System: Project Proposal



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Submitted to:

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1. Introduction and Problem Statement

Project Overview: The Hospital Management System (HMS) is a C++ software solution designed to digitize healthcare operations, including patient records, doctor schedules, and appointment tracking.

The Current Issue: Many facilities still rely on paper-based filing or fragmented spreadsheets.

Consequences of Manual Systems: These outdated methods lead to misplaced patient history, double-booked appointments, and slow retrieval of emergency data.

Stakeholder Impact: Medical staff face high administrative burdens, while patients suffer from longer wait times and risks associated with inaccurate record management.

2. Project Objectives and Scope

Primary Goals: The project aims to create a centralized database, automate scheduling, and provide secure storage for medical and billing history.

Administrative Efficiency: The system allows administrators to generate daily activity reports quickly.

Inclusions: The scope covers patient registration, doctor profile management, appointment booking, and basic billing.

Exclusions: This version does not include real-time surgical monitoring, pharmaceutical stock management, or external laboratory hardware integration.

3. Proposed Solution and System Features

Development Approach: The system utilizes an Object-Oriented approach to model healthcare entities and processes.

System Workflow: Users log in to register patients and assign them to doctors, with all visit details stored in a persistent file.

Access Control: The system features separate login credentials for Admin and Staff members to ensure security.

Search and Retrieval: A searchable database allows staff to find patients instantly using a unique ID or contact number.

Persistent Storage: Data is saved using C++ file handling (.dat or .txt) so that information is retained after the program closes.

4. Technical Design (OOP Principles)

The system is built using a structured class hierarchy to organize data:

Base Class (Person): Stores shared attributes such as Name, Age, and Gender.

Derived Class (Patient): Inherits from Person;stores Patient ID, disease history, and handles billing methods.

Derived Class (Doctor): Inherits from Person; manages specialization, office hours, and availability status.

Appointment Class: Acts as an association to manage the link between a Patient and a Doctor.

Encapsulation: Sensitive data is protected using private members and public setter/getter methods.

Polymorphism: Virtual functions are used for displayInfo() to show unique details for different user types.

5.Expected Output

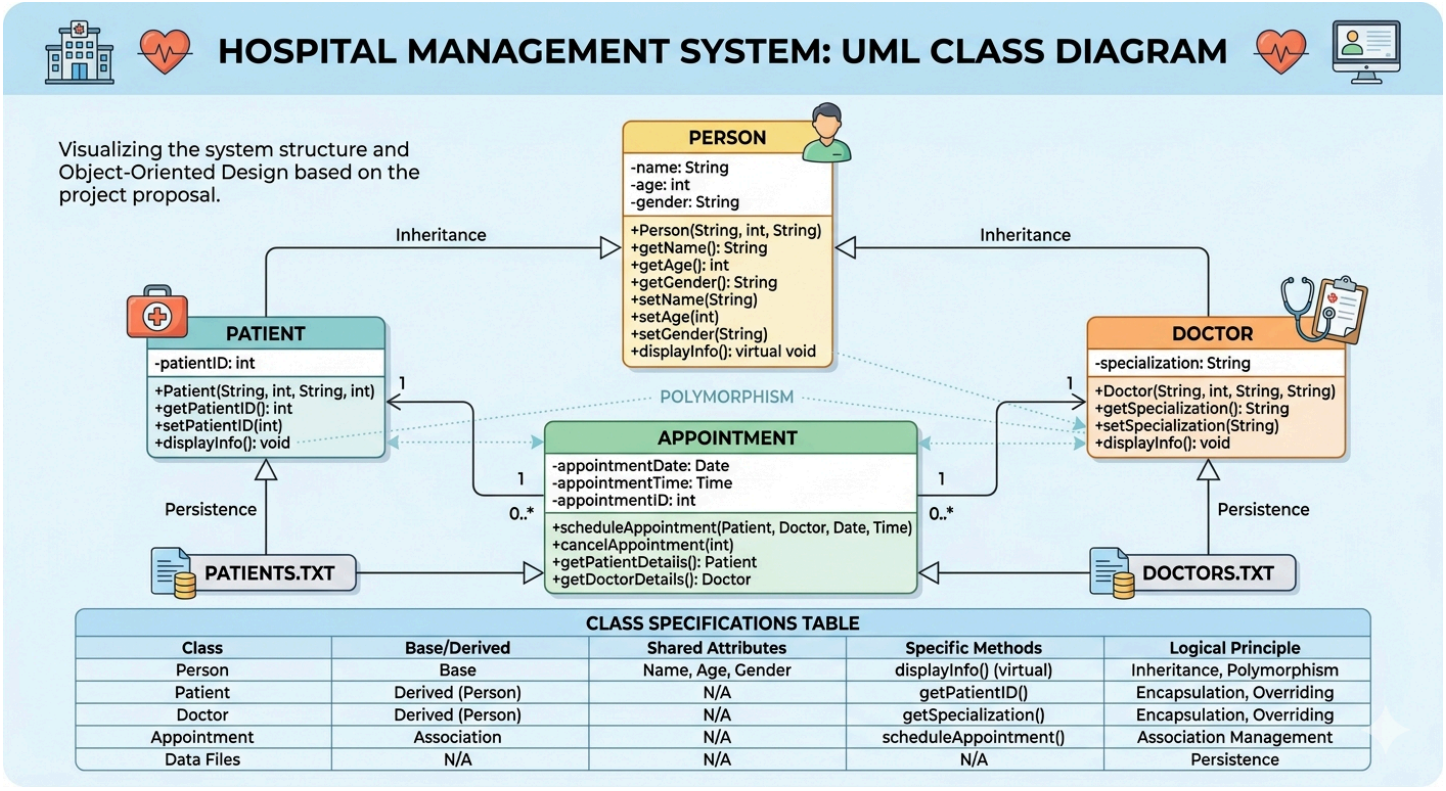
User Interface: A clean, menu-driven console interface where users navigate via numeric inputs.

File Generation: The program automatically creates and maintains patients.txt and doctors.txt files.

Conclusion:

This HMS replaces inefficient manual processes with a modular, easy-to-maintain digital framework capable of handling essential healthcare data with precision.

UML DIAGRAM:



CLASS SPECIFICATIONS TABLE

Class	Base/Derived	Shared Attributes	Specific Methods	Logical Principle
Person	Base	Name, Age, Gender	displayInfo() (virtual)	Inheritance, Polymorphism
Patient	Derived (Person)	N/A	getPatientID()	Encapsulation, Overriding
Doctor	Derived (Person)	N/A	getSpecialization()	Encapsulation, Overriding
Appointment	Association	N/A	scheduleAppointment()	Association Management
Data Files	N/A	N/A	N/A	Persistence